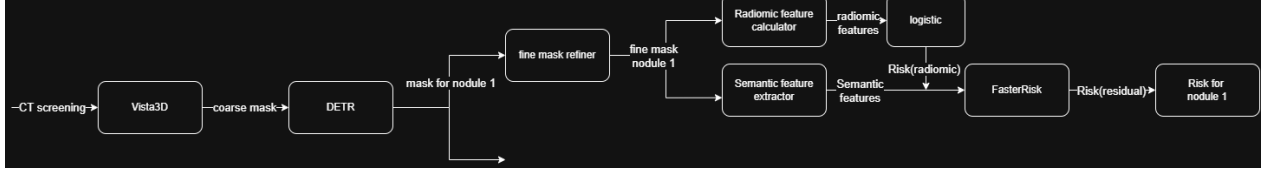


1 Model structure



2 Training objective

V4 combines ten losses:

$$\mathcal{L}_{\text{total}} = \sum_{\ell=1}^{10} \lambda_{\ell} \mathcal{L}_{\ell},$$

where $\mathcal{L}_{\ell}$ is a component loss and λ_{ℓ} is its weight. Table 1 lists the weights used in joint training.

Table 1: Weights of the ten loss components.

Loss component	Weight
Box regression	5.0
Box GIoU	2.0
Coarse Dice	1.0
Coarse focal	1.0
Fine Dice	2.0
Fine focal	1.0
Missing-nodule penalty	2.0
Semantic supervision	1.0
Nodule malignancy	0.5
Scan-level risk	0.5

Averages use eligible training samples; fine-mask losses use only valid ROI voxels. GT denotes ground truth, and ROI denotes region of interest. Superscripts c and f indicate coarse and fine resolution.

3 Loss components

3.1 Box regression

$$\mathcal{L}_{\text{box}} = \text{mean}[|\hat{\mathbf{b}} - \mathbf{b}|].$$

Here $\hat{\mathbf{b}}$ and $\mathbf{b}$ contain the predicted and GT box centers and side lengths, normalized by the scan dimensions. The mean is taken over matched nodules and their six coordinates. This loss encourages accurate nodule location and size.

3.2 Box GIoU

$$\mathcal{L}_{\text{GIoU}} = \text{mean}[1 - \text{GIoU}(\hat{B}, B)].$$

Here $\hat{B}$ and B are matched predicted and GT boxes. GIoU measures their overlap and penalizes unused space in their smallest enclosing box. Averaged across nodules, this loss encourages spatial agreement even when the boxes initially do not overlap.

3.3 Coarse Dice

$$\mathcal{L}_{\text{Dice}}^c = \text{mean} \left[1 - \frac{2 \sum_u p_u g_u}{\sum_u p_u + \sum_u g_u} \right].$$

Here p_u is the predicted coarse foreground probability and g_u is the binary GT mask at voxel u . The sums cover the coarse grid, and the mean covers matched nodules. This loss rewards overall foreground overlap without letting the large background dominate.

3.4 Coarse focal

$$\mathcal{L}_{\text{focal}}^c = \text{mean}[-\alpha_t(1 - p_t)^2 \log p_t].$$

Here p_t is the probability assigned to the correct voxel label: p for foreground and $1 - p$ for background. The class weight α_t is 0.25 for foreground and 0.75 for background. The mean covers voxels in matched coarse masks. The factor $(1 - p_t)^2$ emphasizes difficult voxels and reduces the influence of easy predictions.

3.5 Fine Dice

$$\mathcal{L}_{\text{Dice}}^f = \text{mean}[1 - \text{Dice}(p^f, g^f)].$$

Here p^f and g^f are the refined mask probabilities and cropped GT mask; Dice is the same overlap ratio used for coarse segmentation. It is evaluated on valid voxels and averaged over matched ROIs with sufficient GT coverage. This loss encourages accurate local nodule shape and volume.

3.6 Fine focal

$$\mathcal{L}_{\text{focal}}^f = \text{FL}_+ + 0.25 \text{FL}_-.$$

Here FL_+ and FL_- are the focal expression above, averaged separately over valid voxels in matched positive ROIs and unmatched hard-negative ROIs. Hard negatives receive all-background targets. This loss refines difficult local details and suppresses spurious masks in false candidates.

3.7 Missing-nodule penalty

$$\mathcal{L}_{\text{miss}} = -\text{mean}[\log(oR_{\text{fg}})], \quad R_{\text{fg}} = \frac{\sum_u g_u p_u}{\sum_u g_u}.$$

Here o is the matched query’s object probability, p_u is its coarse mask probability, and g_u is the binary GT mask. The soft recall R_{fg} measures coverage of GT foreground. The mean includes every retained GT nodule, independently of fine-ROI coverage. This loss penalizes low detection confidence and missed foreground, providing an explicit signal against missing nodules.

3.8 Semantic supervision

$$\mathcal{L}_{\text{sem}} = \text{mean}_{i,j} \left[-\sum_{k=1}^5 h_{ijk} \log \pi_{ijk} \right].$$

Here i indexes a valid matched nodule, j a semantic attribute, and k a rating from one to five. The target h_{ijk} is the fraction of readers assigning rating k , and π_{ijk} is the predicted probability. The six attributes are lobulation, margin, sphericity, spiculation, subtlety, and texture. This loss learns interpretable attributes while preserving disagreement among readers.

3.9 Nodule malignancy

$$\mathcal{L}_{\text{nod}} = \text{mean}_{i,m} [\text{BCE}(p_{im}, y_i)], \quad p_{im} = \sigma(a_i^{\text{rad}} + c_{im}).$$

Here i indexes a valid matched nodule and m a selected FasterRisk model. The radiomics baseline logit is a_i^{rad} , the semantic logit correction is c_{im} , and σ is the sigmoid function. The combined malignancy probability p_{im} is compared with the binary GT label y_i using binary cross-entropy (BCE). With the baseline fixed during backpropagation, this loss trains semantic features to explain malignancy beyond radiomics.

3.10 Scan-level risk

$$\mathcal{L}_{\text{scan}} = \text{mean}_m [\text{BCE}(P_m, Y)], \quad P_m = 1 - \prod_i (1 - o_i p_{im}).$$

Here o_i is candidate i ’s object probability and p_{im} its combined malignancy probability under model m . The product runs over all refined candidates. The resulting scan probability P_m is compared with the scan label Y , which indicates whether any retained nodule is malignant. This loss connects detection and nodule risk to the scan-level outcome.

4 Training stages and loss weighting

V4 begins with three warmup epochs, during which MedicalNet is trained on GT-masked nodules using the semantic loss with unit weight. The radiomics baseline and semantic risk models are fitted separately using training annotations. Whole-CT joint training then proceeds for 18 epochs. All ten loss components enter the joint objective at epoch 1 with the fixed weights in Table 1. In particular, the nodule and scan-level losses each have weight 0.5 throughout joint training.

Table 2: Training stages and the activation of loss terms.

Stage	Loss weighting	Optimization
Warmup (3 epochs)	Semantic loss: 1.0	Train MedicalNet on GT-masked ROIs; initialize the radiomics baseline and semantic risk models from annotations.
Joint epochs 1–15	All ten weights in Table 1	Jointly optimize detection, fine segmentation, semantic prediction, and malignancy prediction.
Joint epochs 16–18	All ten weights in Table 1	Additionally unfreeze the last VISTA3D automatic decoder block and its class head.

The training curriculum progressively changes ROI selection and the trainable network layers. The scheduled probability of using GT-centered positive ROIs is 1.0 during joint epochs 1–5 and decreases linearly to zero at epoch 18, increasing reliance on predicted nodule locations. Learning rates follow cosine decay to 10% of their initial values, while the mask temperature remains fixed at one.